DairyGuard: Milk Collection Center Quality Control and Cold-Chain Spoilage Prevention System

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Problem Statement

Dairy farming is a cornerstone of Rwanda's rural agricultural economy, supporting thousands of smallholder households that deliver raw milk daily to local Milk Collection Centers (MCCs). However, MCCs and dairy cooperatives suffer substantial post-harvest losses—estimated between 15% and 20% of daily collected volume—due to milk spoilage, water adulteration, and inadequate cold-chain monitoring. Currently, many MCCs rely on manual paper ledgers to record morning and evening milk deliveries, lactometer density readings, and monthly farmer payments. Without automated validation at the point of collection, warm milk (above 7°C) or diluted, high-acidity batches are frequently poured into bulk cooling tanks, cross-contaminating thousands of liters of good milk, causing tank overflows, and leading to disputed or delayed farmer payouts.

Proposed Solution

DairyGuard is a centralized, normalized (Third Normal Form – 3NF) Oracle relational database powered by PL/SQL business logic to automate Milk Collection Center quality assurance, cold-tank inventory, and farmer payouts. When a farmer delivers milk to an MCC, the technician logs the batch volume along with rapid lab measurements (temperature, pH, lactometer density, and fat content). Oracle PL/SQL triggers and stored procedures immediately validate these metrics before permitting bulk tank assignment—automatically rejecting spoiled or adulterated milk and logging the incident. Furthermore, a modular PL/SQL package automates monthly farmer payouts using quality-tiered pricing (rewarding Grade A deliveries), while security triggers maintain an immutable audit trail of any manual adjustments to test results or financial records.

Data Source

To populate, test, and validate the database with authentic measurements, DairyGuard integrates real laboratory dairy data with Rwandan cooperative operational parameters:

- Primary Dataset: Milk Quality Prediction Dataset sourced from Kaggle (cpluzshrijayan/milkquality), comprising 1,059 records of milk pH, temperature (°C), taste, odor, fat content, turbidity, color, and quality classification.
- Contextual Benchmark Data: National Institute of Statistics of Rwanda (NISR) livestock and dairy production statistics, used to model realistic cooperative collection volumes, tank capacities, and Rwandan Franc (RWF) per-liter payout tiers.
- Licensing & Privacy: The dataset is publicly available under an open educational license and contains zero personally identifiable information (PII). All farmer and cooperative identifiers in the database are anonymized synthetic records.

Planned Tables (3NF Schema)

1. **DISTRICTS_COOPERATIVES:** Stores cooperative metadata, district/sector location, and MCC manager contacts (coop_id PK).
2. **FARMERS:** Stores registered smallholder dairy farmers, herd sizes, cooperative affiliation, and active status (farmer_id PK, coop_id FK).
3. **COOLING_TANKS:** Tracks bulk chiller tanks at each MCC, maximum liter capacity, current stored volume, and operating temperature (tank_id PK, coop_id FK).
4. **MILK_DELIVERIES:** Records daily morning and evening milk deliveries, batch volume in liters, delivery timestamp, and acceptance status (delivery_id PK, farmer_id FK, tank_id FK).
5. **QUALITY_TESTS:** Stores laboratory test readings per delivery—temperature (°C), pH, density, fat percentage, and computed quality grade (test_id PK, delivery_id FK).
6. **FARMER_PAYOUTS:** Tracks monthly billing cycles, total accepted liters, Grade A quality bonuses, net payout amount, and payment status (payout_id PK, farmer_id FK).
7. **AUDIT_LOG:** Captures security events, rejected delivery attempts, unauthorized DML operations, and payout modifications (audit_id PK).

Planned PL/SQL Features

- Triggers:
  - trg_validate_milk_quality: A BEFORE INSERT OR UPDATE trigger evaluates temperature, pH, and density; it blocks cooling-tank assignment and flags a batch as REJECTED if temperature exceeds 7°C or pH falls outside 6.5–6.8.
  - trg_check_tank_capacity: Prevents milk allocation to a COOLING_TANKS record if current_volume + batch_volume exceeds max_capacity.
  - trg_audit_payout_changes: Logs every UPDATE or DELETE operation on FARMER_PAYOUTS and QUALITY_TESTS into AUDIT_LOG with user, timestamp, and old/new values.
- Functions & Procedures:
  - fn_calculate_quality_grade(p_temp, p_ph, p_fat, p_turbidity): Evaluates lab metrics against dairy standards and returns GRADE_A, GRADE_B, or REJECTED.
  - proc_record_delivery(...): Validates active farmer membership, invokes quality grading, assigns an available cooling tank, and updates tank inventory within a single atomic transaction.
  - proc_generate_monthly_payouts(p_month, p_year): Uses an explicit cursor to iterate across active farmers, aggregate accepted deliveries, apply Grade A bonuses, and generate monthly payout records.
- Packages & Exception Handling: pkg_dairy_operations encapsulates collection, quality grading, tank management, and payout routines, implementing custom exceptions (e_contaminated_milk, e_tank_full, e_inactive_farmer) and structured error logging.

Innovation

- Selected Option: Analytics Dashboard (live connection to the Oracle Pluggable Database).
- Tool: Microsoft Power BI connected directly to the Oracle PDB via Oracle ODAC/ODBC drivers.

- Value Proposition: While Oracle PL/SQL enforces quality-rejection rules, tank limits, and audit controls at the database layer, the live Power BI dashboard gives MCC managers and cooperative leaders an interactive visual interface. Managers can monitor daily accepted versus rejected milk volumes, identify spoilage hotspots by district or delivery session, track real-time cooling tank utilization, and evaluate farmer quality-consistency trends.

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